

Three Phones, Nine Afternoons: An Experience Report on Multi-Operator 5G Walk-Testing in Manhattan

Wonyul Choi

AMSE, Morris Hills, NJ

Ahan Kak

Nokia Bell Labs, Murray Hill, NJ

Abstract

Multi-operator cellular field measurements are a foundational tool for understanding commercial networks, yet the published record largely reports cross-operator results while omitting the methodological pitfalls, telemetry artifacts, and silent failure modes that make those results trustworthy; in our experience, such artifacts are the norm, not the exception. We present a methodology-and-experience report from a nine-day weekend walk-test campaign along a fixed ~6 km loop in Midtown Manhattan, instrumenting three U.S. operators simultaneously on identical hardware across paired downlink, uplink, and latency phases. Our central argument is that the hard part of such campaigns is not collecting the data but rather trusting it. We contribute a taxonomy of fourteen measurement artifacts, each with a detection criterion, a mitigation, and where possible a quantified impact. We also present a case study of an NR Secondary Cell Group silent failure in which server-side ground truth and L3 RRC decoding localize a fault that UE-side telemetry alone leaves ambiguous. Additionally, we explain cross-operator throughput, latency-tail, and handover behavior through link-layer metrics. Finally, we provide a supplementary section covering a pre-transition snapshot of a fourth carrier's native standalone network that no longer exists.

ACM Reference Format:

Wonyul Choi and Ahan Kak. 2026. Three Phones, Nine Afternoons: An Experience Report on Multi-Operator 5G Walk-Testing in Manhattan. In *20th ACM Workshop on Wireless Network Testbeds, Experimental evaluation & Characterization (WiNTECH '26)*, October 26–30, 2026, Austin, TX, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3831662.3844164>

1 Introduction

Multi-operator cellular field measurements have long served as a foundational tool for understanding the behavior of commercial wireless networks under real-world conditions [7,

15]. As 5G deployments mature and the wireless community looks ahead to 5G-Advanced and 6G, the value of such measurements continues to grow as they capture configuration choices, deployment patterns, and link-layer behaviors that no laboratory testbed can reliably reproduce. Consequently, an expanding body of work in the measurement community now quantifies cross-operator performance differences across the radio, transport, and application layers [2, 4, 5, 12]. However, the existing literature typically does not address methodological pitfalls, telemetry artifacts, and silent failure modes, all of which complicate the analysis of real-world data. In our experience conducting multi-operator walk-test campaigns, these artifacts are not the exception but the norm. For example, RAN infrastructure vendor-specific behavior exposes different events for different operators on identical hardware; UEs occasionally report nominal attachment while silently failing to carry user-plane data over the intended bearer, among other cases. Despite long-standing calls for sound measurement practice and the documentation of calibration issues [10], the artifacts themselves, along with the associated detection criteria, mitigation strategies, and quantified impacts, rarely make it into the published record. As a result, researchers conducting similar campaigns must rediscover them. Through this paper, we aim to close that gap. We present a methodology-and-experience report grounded in a nine-day weekend-afternoon walk-test campaign¹ along a fixed ~6 km loop in Midtown Manhattan. The campaign uses three Motorola Edge 2024 UEs, each running Network Signal Guru (NSG) [11], an Android-native modem diagnostic tool with native JSON export, to simultaneously instrument three U.S. mobile operators² across paired downlink (DL) and uplink (UL) iperf3 phases and parallel round-trip-latency probes. A supplementary two-day pilot dataset additionally captures a U.S. “fourth carrier” on its native 5G standalone (SA) infrastructure before that network was wound down, alongside the same UE on its post-transition shared radio access network (RAN) configuration. More specifically, through this paper, we provide the following contributions.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

WiNTECH '26, Austin, TX, USA

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2879-2/26/10

<https://doi.org/10.1145/3831662.3844164>

¹Campaign Dataset: <https://github.com/wchoi27/acm-wintech-2026-manhattan-cellular-measurement>.

²Throughout the paper, operators are anonymized as Carrier A, Carrier B, and Carrier C, with relative ordering on cross-operator metrics preserved without naming. Additionally, the “fourth carrier” is labeled Carrier D.

- A paired three-UE measurement framework for simultaneous multi-operator instrumentation, evolved across a nine-day campaign (Section 3).
- A taxonomy of fourteen measurement artifacts, with detection criteria, mitigations, and quantified impact, and a case study that distinguishes UE-side failures from server-side ones (Sections 5 and 6).
- A link-layer explanation of cross-operator throughput differences via NR carrier aggregation, MIMO rank, modulation, block error rate (BLER), and channel quality (reference signal received power and quality, i.e., RSRP and RSRQ) (Section 7).
- A pre-transition snapshot of the U.S. “fourth carrier” on its native 5G SA infrastructure (Appendix A), documenting a network that no longer exists in that form.

2 Background and Related Work

We position our work along three axes: multi-operator 5G field measurement, low-level telemetry tooling, and the U.S. “fourth-carrier” deployment context.

Multi-Operator 5G Measurement. A substantial body of work characterizes commercial 5G performance, mostly via vehicular drive-tests with professional tools, quantifying cross-operator throughput, coverage, and latency and the lower-layer choices that explain them (spectrum, frame structure, carrier aggregation, RB utilization) [2, 12]; recent attention has turned to standalone (SA) deployments [4]. Our work differs in three ways. First, our data is collected on foot rather than from a vehicle, sampling the radio environment at a spatial granularity unavailable to drive-tests and exposing the dense-handover behavior characteristic of urban canyons (Section 7). Second, we instrument three operators *simultaneously* on identical hardware, enabling a within-session paired comparison that ensures identical time, weather, and walker variables (Section 3). Third, and most central to this paper, our contribution is not the cross-operator ranking itself but the methodological scaffolding required to obtain it reliably, i.e., the artifacts, failure modes, and corrections that the results-focused measurement literature does not typically report [10]. The closest prior art within this domain characterizes coverage of 5G non-standalone (NSA) deployments on a single operator [8]; we extend that line to three operators measured concurrently and foreground the experience-report dimension over coverage characterization alone.

Cellular Telemetry Tooling. Extracting below-IP-layer cellular information from commodity devices is well established. MobileInsight pioneered on-device decoding of 3GPP Layer-3 signaling on smartphones [9]; a complementary line places telemetry at the network edge, with NG-Scope decoding LTE physical-layer downlink control information [14] and NR-Scope extending this passive, base-station-side approach

to 5G SA [13]. We occupy a different point in this design space, using Network Signal Guru (NSG), an Android-native tool that records and decodes the full protocol stack from physical-layer measurements through Layer-2 to Layer-3 RRC/NAS signaling. Widely used in industry RF engineering but under-represented in research, NSG motivates a secondary contribution: documenting what its output does and does not expose, and the pitfalls that follow (Section 5).

The U.S. “Fourth-Carrier” Context. The emergence of a fourth U.S. mobile network operator produced a short-lived native 5G SA deployment that has since been substantially restructured: the operator transitioned its traffic onto a network-sharing arrangement with another carrier, winding down portions of its own RAN. RAN sharing of this kind has itself become a subject of recent measurement study [3]. To our knowledge, however, no research field-measurement record exists of the fourth carrier’s *native* deployment prior to the transition. Our supplementary section (Appendix A) presents a snapshot of the native SA network in its final weeks of operation, alongside post-transition measurements of the same device on the shared-RAN configuration.

3 The Campaign as It Happened

The Setup. The measurement setup consists of three Motorola Edge 2024 (Snapdragon X62) handsets, mounted in a three-phone chest harness so that all three traverse an identical trajectory at the same instant. Each handset runs NSG and carries the SIM of one operator; the operator-to-slot assignment is rotated between walk-days to balance any harness-position bias. The walker proceeds at a natural pedestrian pace of ~ 1.3 m/s. On each handset, an *iperf3* client is orchestrated by a companion application, while the matching *iperf3* server and the latency reflector are co-located on a single fixed, publicly reachable host per operator, with all three hosts located within the same datacenter. The host was deliberately sited far from the test corridor, on the order of a cross-country path, so that the latency measurements reflect a realistic wide-area round trip rather than a co-located best case; the observed round-trip-time (RTT) floor is ~ 35 ms (Section 7). Latency is probed with the Two-Way Active Measurement Protocol (TWAMP) [6] using an open-source Android client and Linux reflector [1].

The Loop. As shown in Figure 1, the route is a fixed ~ 6 km loop through Midtown Manhattan, spanning roughly 42nd to 59th Streets (a north-south extent of ~ 1.6 km) between the avenues bounding the corridor on its east and west sides.

Framework Evolution Across Nine Days. The measurement framework was not fixed in advance and evolved across the nine days as the field demanded, as documented in Figure 2. By March 22 it had settled into the baseline we retained as the campaign spine: a line-rate DL phase, a line-rate UL phase, and a TWAMP latency phase. Several days departed

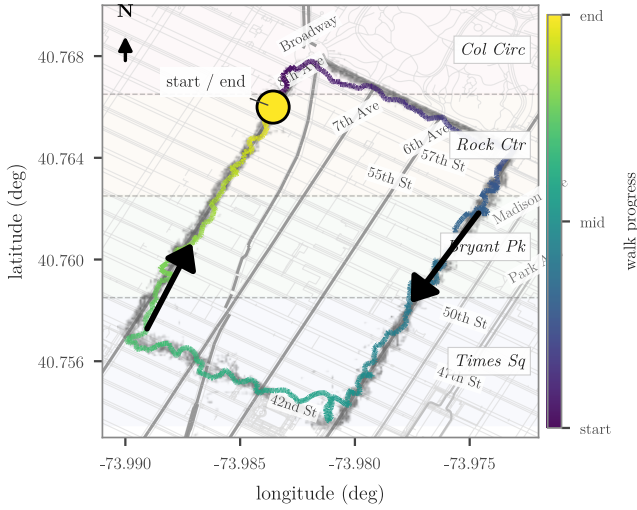


Figure 1: The fixed ~6 km Midtown loop over the underlying street grid.

from it instructively. On March 29, Carrier C returned no TWAMP responses. On April 25 we added server-side `iperf3` logging (*Server-Log Integration* below); this is also the day Carrier A’s NR Secondary Cell Group (SCG) silently stopped carrying data (Section 6), persisting into April 26 when Carrier B also showed a partial NR primary-cell drop on DL. Finally, on May 2 and 10 we introduced the rate-capped regime (*Three Traffic Regimes* below), replacing many short line-rate sessions with one long rate-limited session per phase.

Three Traffic Regimes. The campaign spans three traffic regimes, each interrogating a different operating point. The *line-rate* regime drives `iperf3` over UDP at a target rate in excess of typical channel capacity (1000 Mbps and above); here the wireless channel is the bottleneck, substantial packet loss is expected, and the loss and throughput that result are informative about channel capacity. The *rate-capped* regime drives UDP at a target well below channel capacity (20–50 Mbps); here the configured cap is the bottleneck, near-zero loss is expected, and the regime probes how the networks behave when the offered load is modest rather than saturating. The *latency* regime uses TWAMP [6] probes against the fixed remote reflector.

Multi-Operator Simultaneity and Pairing. Because the three handsets share a harness and traverse the route together, any comparison *within* a simultaneous session is paired: the operators experience the same instant, weather, walker, and urban-canyon geometry. This pairing underlies the cross-operator claims in Section 7 and removes the time-of-day, route, and environmental variations associated with sequential single-operator measurement. We are correspondingly careful across this boundary, reporting per-walk

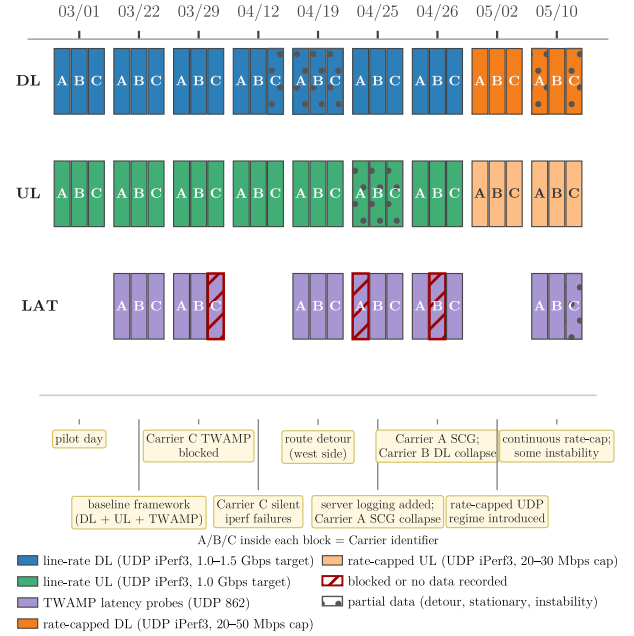


Figure 2: Framework evolution across the nine walk-days.

distributions rather than pooling cross-session or cross-day data indiscriminately.

Server-Log Integration. From April 25 onward we logged the server side of every `iperf3` session, aligning each server-side connection-accept with its NSG test-case record one-to-one. This lets the server’s view of offered and delivered bitrate be cross-referenced against the UE’s physical-, MAC-, and PDCP-layer throughput, which is what distinguishes a radio-side anomaly from a server-side one and is essential to the case study in Section 6.

4 Dataset Overview

After filtering to walks exceeding 5 MB and lasting 30 minutes, the campaign comprises 83 walk sessions across the nine days, containing 2,443 `iperf3` test cases. On the four server-logged days, 603 connection accepts align one-to-one with the UE record. The rate-capped phase uses fewer but longer sessions (9 and 7 versus 273 and 314 on the line-rate days). Of 17 TWAMP tests, 16 carry fully intact summaries, while one on May 10 is corrected for a corrupted packet count. Table 1 characterizes the per-operator deployment as observed in the data. The deployment contrast is readily visible here and motivates the findings that follow. Carriers A and B operate non-standalone, pairing a single mid-band NR carrier with a multi-band LTE anchor, whereas Carrier C operates standalone with NR carrier aggregation and only a minimal LTE presence. The cell-count rows foreshadow a second contrast: Carrier A observes far more distinct LTE

Table 1: Per-operator deployment as observed in the dataset (anomaly days excluded from cell counts).

	Carrier A	Carrier B	Carrier C
Network mode	NSA	NSA	SA
Primary NR band	n77	n77	n41
NR bandwidth	80 MHz	100 MHz	100 MHz
NR carriers (mean)	1	1	1–3 (~2.2)
LTE band count	6	4	minimal
LTE cells observed	2,046	1,135	148
LTE cells camped	567	254	49
NR cells observed	476	437	678
NR cells camped	89	84	105

cells (2,046) than it ever uses (567), while Carrier C’s footprint is concentrated on the NR side, where it both observes and camps on more cells than either NSA operator (Section 7). Although the route is fixed, GPS does not trace it identically on every walk. Across the campaign, only ~1.6% of samples (4,859 of 297,578) fall more than ~5 m off the canonical footprint, and these deviations concentrate on a handful of days rather than spreading evenly, so the route is consistent enough that cross-day spatial comparison is meaningful.

5 A Taxonomy of Measurement Artifacts

A recurring lesson of the campaign is that raw telemetry cannot be trusted at face value: across nine days we cataloged fourteen artifacts, each of which would distort a downstream analysis if left undetected. Table 2 summarizes them with a detection criterion and a mitigation; they fall into three groups: operator and field-procedure effects (A1–A4), telemetry-tool semantics (A5–A7, A9–A14), and one cross-cutting failure mode (A8, Section 6). Four artifacts are specifically relevant to the results that follow and warrant a note here. A7: NSG’s `RB_Num_Average_DL` is a *per-carrier* mean, not a cross-carrier sum, so a throughput model must form an effective RB count as the per-carrier mean times the number of serving NR carriers, or it under-credits multi-carrier operators. A9: in MR-DC (NSA) mode NSG reports a combined NR+LTE `MRDC_*_Throughput` field that is absent for SA operators (whose value lives in `Physical_Throughput`) and reading one field uniformly zeroes the SA carrier, so the pipeline must branch on mode. A13: NSG labels the NR serving-cell with a `PCell_*` prefix in both modes, which is the PSCell under NSA and the PCell under SA, therefore we use the neutral term “NR serving-cell” where relevant. A14: an aggregate “handover count” mixes NSA PSCell changes with SA full PCell handovers, which have different RRC-layer latencies, so we classify each by its PDU contents and report counts and latencies per class (Section 7).

Table 2: The fourteen measurement artifacts: detection criterion and mitigation.

Artifact	Detection	Mitigation
A1. Latency measurement traffic dropped on probe port	100% TWAMP loss vs. normal connectivity	Run a second latency probe (ICMP/HTTP) in parallel
A2. Stationary test clusters bias spatial bins	Per-test GPS box, e.g., <~20 m	Flag; exclude from per-location stats, keep for per-walk
A3. Route variation thins per-location coverage	Samples outside 9-day occupancy grid (~1.6%)	Keep for per-walk; exclude from per-location bins
A4. Identical HW exposes different telemetry per operator	Vendor RRC event types differ across operators on same UE	Compare only fields all operators populate
A5. WCDMA-related telemetry but no WCDMA networks	WCDMA record has NAS-only sub-block, no PCI/radio	Require a RAT-specific radio/PCI field before counting
A6. Tool emits vendor-specific non-UTF-8 bytes in JSON, impacting decode	<code>json.load</code> raises <code>UnicodeDecodeError</code>	Decode with <code>errors='replace'</code> ; only opaque IMS string affected
A7. RB field is a per-carrier mean, not an aggregate	Multi-carrier RB count falls below primary cell’s own	Effective RB = per-carrier mean × NR carriers
A8. NSA NR-SCG silent failure	NR serving-cell presence <~5% despite normal LTE	Per-walk NR-presence check; flag, exclude, re-run (Section 6)
A9. NSA-only throughput fields read zero for SA	<code>MRDC_*</code> field absent for SA carrier	Branch on mode; read <code>Physical_Throughput</code> for SA
A10. For some operators, modem clock is on GPS time, not UTC	Per-walk median (host–modem) timestamp ≈ 18 s	Use host timestamp for all cross-tool joins
A11. First record carries a 1980 epoch	Modem timestamp reads Jan. 1980 (GPS epoch)	Drop pre-lock records at parser stage
A12. One RSRQ field is measured, the other is derived	One field equals SSB-RSRQ, another exactly 3 dB below	Treat as redundant; report the measured SSB value
A13. NR “PCell” field is the PSCell under NSA	<code>PCell_*</code> prefix used in both NSA and SA	Read as PSCell (NSA)/PCell (SA); use “NR serving-cell”
A14. “Handover count” mixes distinct procedures	NSA PSCell change vs. SA full PCell HO conflated	Classify by PDU; report counts/latencies per class

6 Case Study: The NR-SCG Silent Failure

We reproduce the most instructive artifact (A8) as a case study, because it exercises every methodological tool the paper advocates: per-walk anomaly detection, server-side ground truth, and L3 RRC root-cause analysis. The canonical instance is Carrier A’s April 25 download walk.

Table 3: RRC signaling on a healthy versus a failed Carrier A download walk.

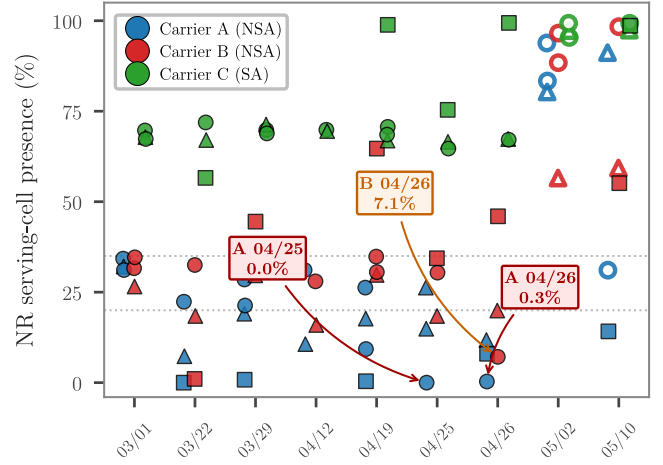
Signal	Healthy (04-12)	Failed (04-25)
LTE reconfigs		
carrying NR meas. config	250	85
SCG-addition triggers	197	0
NR RRC Reconfiguration PDUs	772	0
NR RACH attempts	114	0
NR RACH successes	114	0

Empirical Signature. The UE is attached in NSA EN-DC, but the NR serving-cell sub-block is absent from virtually all records (0.0% on April 25, 0.3% on April 26, against a 22–34% baseline). NR PDCP throughput is zero across all 3,324 records of the walk. The LTE anchor reports normal coverage (RSRP ~ -78.7 dBm, marginally below the typical -74 dBm but well within working range), yet this anchor delivers only ~ 1.7 Mbps median, far below its usual multi-carrier tens of Mbps. The failure is spatially uniform across the entire loop. Other operators were unaffected, save for a separate, smaller Carrier B download-only PSCell drop on April 26.

Server-Side Evidence Localizes the Fault to the UE. The Carrier A server log for April 25 shows the iperf3 server accepting its connections and emitting at the full target rate throughout the download phase. However, the UE received almost nothing. Without this server-side record, the same UE-side signature (zero NR throughput, near-zero NR serving-cell presence) could equally have arisen from a server-side iperf3 failure. The field-log integration is therefore what makes the diagnosis possible.

RRC Signaling Localizes the Fault to the NR Measurement Layer. NSG exposes the full L3 RRC PDUs. Table 3 compares the failed walk against a healthy Carrier A reference (April 12) and we note the following: the network configured the UE to measure NR (the LTE measurement configuration pointed at NR n77 with standard SSB parameters, on both walks), but the UE never returned a qualifying NR measurement report, so the network never issued an SCG-addition reconfiguration; therefore, no NR-direct RRC signaling occurred, and the UE never even attempted physical NR access (zero RACH attempts across a ~ 50 -min. walk).

What This Rules Out. The sequence excludes an SCG addition that failed midway (it was never initiated) and a modem bug suppressing NR scheduling (there was none to suppress). The most specific supportable claim is that the UE never achieved an NR measurement good enough to cross the reporting threshold, placing the fault at the NR measurement and receive layer. Distinguishing “measured but below threshold” from “never measured” would need modem-internal logs that NSG does not export, so we stop at the boundary the data supports.

**Figure 3: NR serving-cell presence rate per walk; marker shape encodes phase (circle DL, triangle UL, square latency).**

Key Takeaways. The signature, a UE that looks attached while the NR serving-cell is silent, with server-side traffic confirmed healthy, is largely independent of the specific UE, modem, and network, and is a pattern other campaigns will encounter. Figure 3 shows NR serving-cell presence per walk across the campaign, isolating the Carrier A outliers alongside the smaller Carrier B April 26 drop. The practical lesson is twofold: first, exclude such walks from typical-performance claims, and second, treat such observations as indicative of failure modes worth exploring further.

7 Headline Measurement Findings

In this section, we present our key findings. Across the campaign, NR is the active RAT 95% of the time on Carrier C and 51% on Carriers A and B, and all cross-operator comparisons exclude the anomaly days of Section 6. These results reflect the experience of a single subscriber on one non-FR2 device over nine weekend afternoons along a fixed route, and are not intended as generalizable, carrier-level performance claims.

Radio Metrics. Figure 4 shows per-operator CDFs of NR and LTE RSRP and RSRQ. Median NR RSRP is -78.2 , -75.0 , and -72.0 dBm for the three operators; median NR RSRQ is -11.9 , -10.9 , and -10.8 dB. The ~ 1 dB RSRQ deficit for Carrier A is also consistent with a noisier or more heavily loaded channel and with the higher block error rate reported below. Per-record Pearson correlation between NR RSRP and RSRQ is only 0.44–0.51, confirming that the two capture distinct aspects of channel state. Figure 4 also plots LTE RSRP/RSRQ, but Carrier C’s minimal LTE presence renders the LTE cross-operator comparison uninformative.

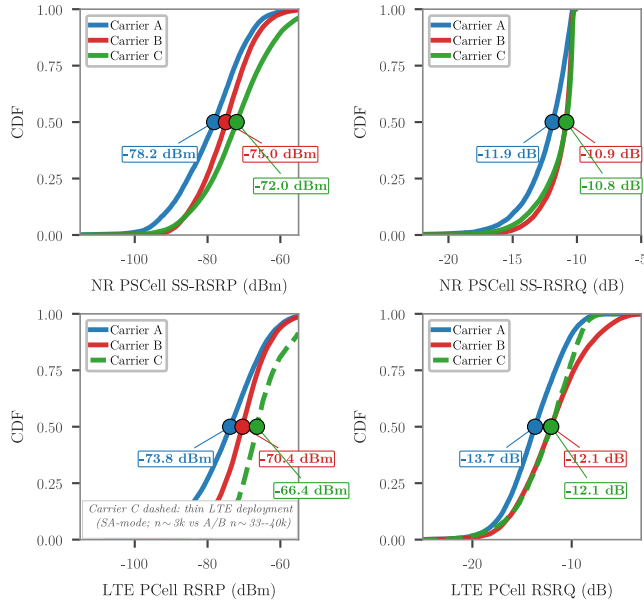


Figure 4: Per-operator CDFs of NR and LTE RSRP and RSRQ over line-rate phases. Markers represent medians.

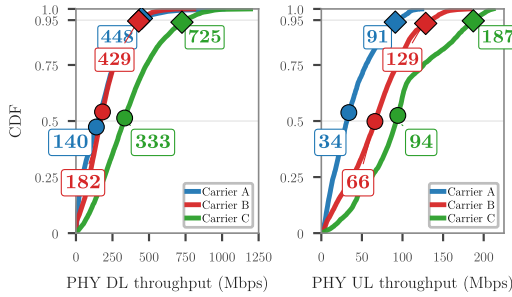


Figure 5: Per-walk physical-layer throughput CDFs, line-rate phases. Circles mark per-walk medians, diamonds the p95.

Throughput Performance. Figure 5 shows per-walk physical layer throughput CDFs. Per-walk median DL throughput is 140, 182, and 333 Mbps for Carriers A, B, and C; UL medians are 34, 66, and 94 Mbps. For SA carriers this physical throughput is the pure NR rate, whereas for NSA carriers it is the MR-DC aggregate (NR plus LTE anchor; artifact A9, Table 2). However, the LTE-anchor contribution to the NSA aggregate is small relative to the NR side. Per-day variance also differs by operator: Carrier A’s DL median ranges 82–190 Mbps across days (a 2.3 $\times$ spread), while Carriers B and C show a smaller variance (177–196 and 283–386 Mbps respectively, excluding the collapse days).

Table 4: NR DL link-layer metrics under active scheduling.

	MCS	BLER	Layers	Carriers	Eff. RB
Carrier A	15.8	22.5%	2.84	1.00	152
Carrier B	13.6	18.0%	2.40	1.00	213
Carrier C	14.2	18.4%	3.33	2.32	312

Link-Layer Cell Performance. The link-layer metrics in Table 4 are NR-only by design: they are per-radio-link quantities measured against the NR serving-cell, since combined NR+LTE link-layer aggregates do not exist in the standard. Carrier C’s throughput is explained almost entirely on the NR side, and in order of contribution: NR carrier aggregation (mean 2.32 carriers under active scheduling), a higher MIMO rank (3.33 layers versus 2.84 and 2.40), and a marginally cleaner channel. The product of per-carrier resource blocks and carrier count, i.e., the effective RB count, is 152, 213, and 312 for the three operators, tracking the throughput ordering. We observe that Carrier A pairs the highest single-carrier modulation (mean MCS 15.8 versus 13.6 and 14.2) with the highest block error rate (mean BLER 22.5% versus 18.0% and 18.4%) and the lowest NR throughput of the NSA pair. The higher modulation and higher block error rate co-occur, consistent with link adaptation operating at a more aggressive point on the rate-reliability curve. The LTE anchor contributes modestly: active-period LTE PDCP DL median is $\sim$ 13 Mbps for Carrier A and $\sim$ 31 Mbps for Carrier B, the latter higher because Carrier B aggregates LTE carriers more often (mean 1.62 versus 1.42) despite a lower modulation order. Figure 6 shows the NR DL modulation mix per operator and per zone. Notably, 1024-QAM is never observed at street level for any operator on any day. The 256-QAM share drops sharply (more than halving, from $\sim$ 13% to $\sim$ 5%) in the Columbus Circle high-rise canyon even where the aggregate MCS holds up, making the modulation mix a more spatially sensitive degradation indicator than MCS alone.

Latency Performance. Figure 7 shows the TWAMP round-trip latency CDFs. Because the reflector sits far from the test corridor, the absolute floor ($\sim$ 35 ms) is dominated by the wide-area path rather than the access network, so the informative comparison is in the tail, where all three operators hit the same reflector simultaneously. Median RTT is close across operators (62.7, 62.4, and 68.6 ms), and the p95 is similar (88–98 ms), but the p99 differs by operator: 225 ms for Carrier A, 291 ms for Carrier B, and 168 ms for Carrier C.

Rate-Capped Regime. When the offered load is capped well below capacity (20–50 Mbps on May 2 and 10), all three operators meet the cap with essentially identical median

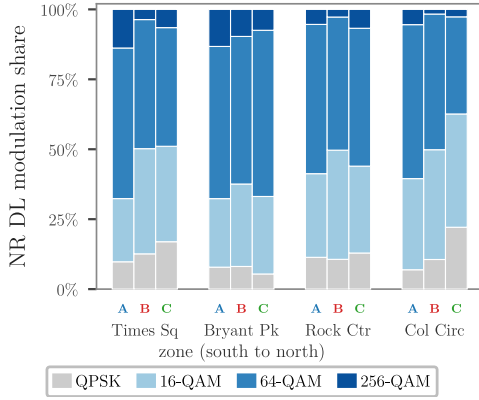


Figure 6: NR DL modulation-order share per operator and per zone, active-scheduling samples.

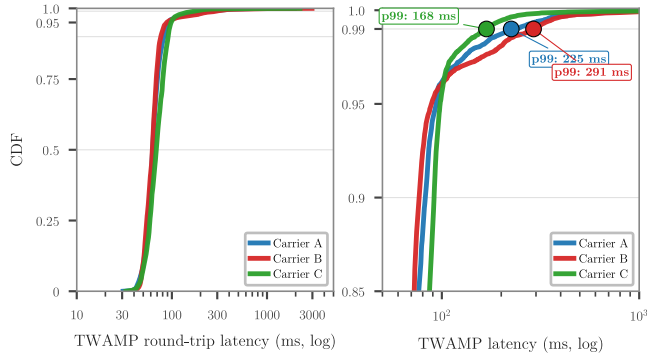


Figure 7: TWAMP round-trip latency: full CDF (left) and tail zoom (right).

throughput (e.g., at a 30 Mbps DL cap, 32/31/30 Mbps; at a 20 Mbps UL cap, 22/24/24 Mbps). The two regimes thus answer different questions: line-rate reveals channel capacity (where operators differ by 2.4 $\times$), rate-capped reveals whether a modest target is met reliably (where they are indistinguishable).

Mobility and Handovers. As artifact A14 (Table 2) notes, counting handovers naively mixes structurally different procedures, so we classify each NR reconfiguration-with-sync by its RRC PDU contents and report counts and RRC-layer interruption times per class (Table 5). We measure interruption time by pairing each `rrcReconfiguration` carrying `reconfigurationWithSync` with its subsequent `rrcReconfigurationComplete`, using NSG’s per-message modem timestamps. The most visible contrast is between Carrier C’s SA intra-NR PCell handover (median $\sim$ 47 ms) and Carrier B’s NSA SCG-change-with-sync (median $\sim$ 13 ms). These are different 3GPP procedures as the SA case is a full primary-cell handover, while the NSA case reconfigures the secondary cell group with the LTE anchor left in place. We report the

Table 5: NR handover events by class, with median RRC-layer interruption time.

Op	Class	Count	Median (ms)
A	SCG change w/ sync (NSA)	545	20
A	SCG release-and-add (NSA)	380	8
A	SCG re-add on		
	LTE PCell HO (NSA)	669	24
B	SCG change w/ sync (NSA)	3,425	13
B	SCG re-add on		
	LTE PCell HO (NSA)	735	43
C	Intra-NR PCell HO (SA)	4,788	47

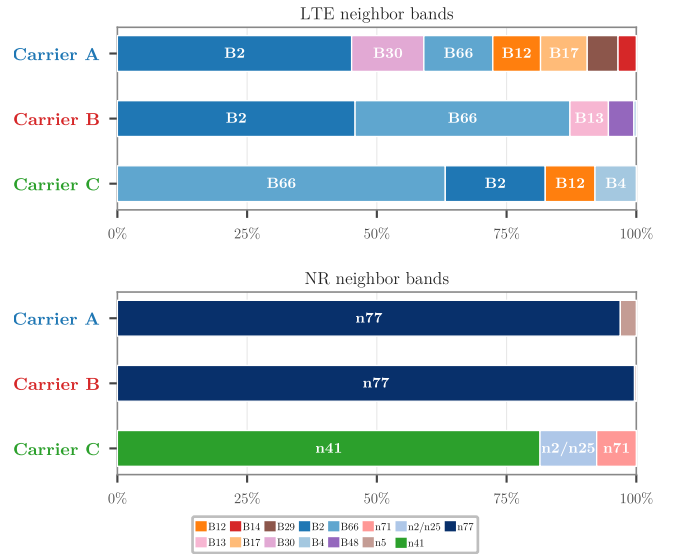


Figure 8: Neighbor-cell band distribution per operator (LTE top, NR bottom).

gap as an empirical observation, not a causal claim: the literature establishes no expected field ratio, and operator-, UE-, and firmware-specific effects cannot be ruled out from these logs. Aggregated per walk, Carrier C makes about 12% more NR handovers than Carrier B (204 versus 182 per walk, mean) but spends roughly 2.5 $\times$ longer in NR RRC reconfiguration per walk (8.9 s versus 3.5 s, against 1.7 s for Carrier A), the combined effect of higher handover frequency and longer per-handover interruption. Even so, this is only 0.25% of walk time, and the latency tail of Figure 7 shows no handover-aligned spikes, so the RRC overhead does not appear to cause bulk application-layer loss in our probes.

Cell-Environment Topology. NSG reports cells in two roles: *observed* (appearing in any measurement report, serving or neighbor) and *camped on* (ever serving the UE). The two diverge because an NSA UE sweeps wide LTE neighbor lists for mobility decisions while using only a fraction of

them, and reporting both separates what the UE sees from what it uses (Table 1). The gap also tracks signal quality as we note that neighbors are systematically weaker than the serving cell (LTE by 5–10 dB, NR by 15–20 dB, consistently across operators), so the wide neighbor footprint is a mobility-measurement perimeter, not a dense alternative coverage layer. Figure 8 shows the neighbor band distribution. A few operator-specific notes follow from it, e.g., one operator deploys CBRS commercially, while another runs three distinct NR bands.

8 Conclusion

We have presented a methodology-and-experience report from a nine-day multi-operator 5G walk-test campaign in Midtown Manhattan, instrumenting three operators simultaneously on identical hardware. Our central argument is that the hard part of such campaigns is not collecting the data but trusting it: across nine afternoons we cataloged fourteen telemetry artifacts and silent failure modes, each of which would corrupt a cross-operator comparison if taken at face value. For each we give a detection criterion, a mitigation, and where possible a quantified impact, and we show how server-side ground truth and L3 RRC decoding localize a failure that UE-side telemetry alone leaves ambiguous. The cross-operator findings our campaign reveals are the payoff of this methodology. We hope the artifact taxonomy and detection-and-mitigation recipes save the next campaign the cost of rediscovering them, and that the dataset lets others test their pipelines against ground we have already mapped.

References

- [1] Murat Demirten. [n. d.]. Twamp-gui: Cross Platform TWAMP Client and Responder with Qt QML Backend. <https://github.com/demirten/twamp-gui>.
- [2] Rostand A. K. Fezeu et al. 2024. Unveiling the 5G Mid-Band Landscape: From Network Deployment to Performance and Application QoE. In *Proceedings of the ACM SIGCOMM 2024 Conference*. ACM, 652–665.
- [3] Rostand A. K. Fezeu et al. 2026. Spectrum and RAN Sharing: A Measurement-Based Case Study of Commercial 5G Networks in Spain. arXiv preprint arXiv:2601.13484.
- [4] Moinak Ghoshal et al. 2025. A First Large-Scale Study of Operational 5G Standalone Networks. *Proceedings of the ACM on Networking* 3, CoNEXT4 (2025), 1–21.
- [5] Ahmad Hassan et al. 2022. Vivisecting Mobility Management in 5G Cellular Networks. In *Proceedings of the ACM SIGCOMM 2022 Conference*. ACM, 86–100.
- [6] K. Hedayat et al. 2008. A Two-Way Active Measurement Protocol (TWAMP). RFC 5357. Internet Engineering Task Force.
- [7] Junxian Huang et al. 2013. An In-Depth Study of LTE: Effect of Network Protocol and Application Behavior on Performance. In *Proceedings of the ACM SIGCOMM 2013 Conference*. ACM, 363–374.
- [8] Konstantinos Kousias et al. 2022. Coverage and Performance Analysis of 5G Non-Standalone Deployments. In *Proceedings of the 16th ACM Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (WiNTECH)*. ACM, 61–68.
- [9] Yuanjie Li et al. 2016. MobileInsight: Extracting and Analyzing Cellular Network Information on Smartphones. In *Proceedings of the 22nd*

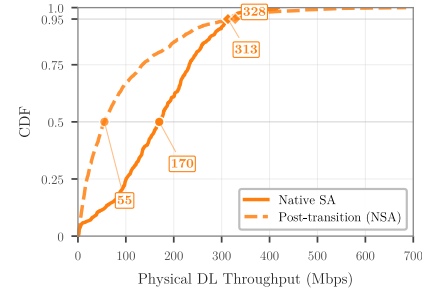


Figure 9: Carrier D physical DL throughput CDFs; circles mark medians, diamonds the p95.

- Annual International Conference on Mobile Computing and Networking (MobiCom)*. ACM, 202–215.
- [10] Vern Paxson. 2004. Strategies for Sound Internet Measurement. In *Proceedings of the 4th ACM SIGCOMM Conference on Internet Measurement (IMC)*. ACM, 263–271.
 - [11] Qtrun Technologies. 2024. Network Signal Guru (NSG). <https://www.qtrun.com>. Cellular drive-test and protocol-analysis tool.
 - [12] Muhammad Iqbal Rochman et al. 2025. A Comprehensive Real-World Evaluation of 5G Improvements Over 4G in Low- and Mid-Bands. *IEEE Transactions on Cognitive Communications and Networking* 11, 3 (2025), 1427–1441.
 - [13] Haoran Wan et al. 2024. NR-Scope: A Practical 5G Standalone Telemetry Tool. In *Proceedings of the 20th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT)*. ACM, 73–80.
 - [14] Yaxiong Xie and Kyle Jamieson. 2022. NG-Scope: Fine-Grained Telemetry for NextG Cellular Networks. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 6, 1 (2022), 1–26.
 - [15] Dongzhu Xu et al. 2020. Understanding Operational 5G: A First Measurement Study on Its Coverage, Performance and Energy Consumption. In *Proceedings of the ACM SIGCOMM 2020 Conference*. ACM, 479–494.

A A Snapshot of the “Fourth Carrier”

A two-day pilot, predating the nine-day framework, captured the fourth carrier of Section 2 (Carrier D) once on its native SA network and once after the transition to shared RAN.

Native SA. On the pre-transition day, Carrier D ran NR-only (SA) on a single n70 mid-band carrier with no LTE anchor, surfacing 66 distinct NR cells. Median NR physical DL throughput was 170 Mbps on one 25 MHz carrier, with a high mean MIMO rank (3.8 DL layers) and the lowest BLER of any operator (12% versus 18–23%), which is consistent with a capable but lightly loaded network where low contention sustains high rank and conservative error rates. We make no scale or coverage claim from a single walk.

Transition to Shared NSA. After the transition, the same UE attached in NSA on the host incumbent’s NR and LTE bands, seeing 125 distinct NR cells. Median DL throughput fell to 55 Mbps, thereby tracking the shared cells’ congestion since peak throughput was unchanged. The measurable direction is thus from a near-empty native network with high median throughput to a shared one whose median reflects the host operator’s load.